

SLM Paper - Initial Sketch

Thoughts

1. Why is deepseek underperforming ?
2. Has the model been fine tuned or will fine tuning perform better ?
3. Does longer solution correlate to good solve ?
4. Are thinking models generally underperforming ?
5. Does deepseek lack a proper parser in the pipeline ?
6. Why do we need Semantic Consistency here ?
7. What if we implement Langchain Sequential chain ?

1

- Use phi-14b dataset with their exact methods to reproduce the results

2

- Test to see if deepseek having the same issue

3

- Try to parse deepseek differently

4

- Try Agentic method combining thinking and coding models

5

- Use distilled models of LLMs ?

6

- Try Error correction by recursive method ?

Work done till now :

1. Reproduced the results (Phi - 14B - 1 Language)
2. Trying Changin Prompt for Deepseek —(Worsens Score)
3. Trying to utilize thinking models to prune syntax errors after generating (WIP)